

SimpleNet: A Simple Network for Image Anomaly Detection and Localization

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Problem & Motivation

What is Anomaly Detection?

- * Goal : Identify deviations from normal patterns
- * Constraint : Trained on Normal Data only (One-Class Learning)

Why is it Hard ?

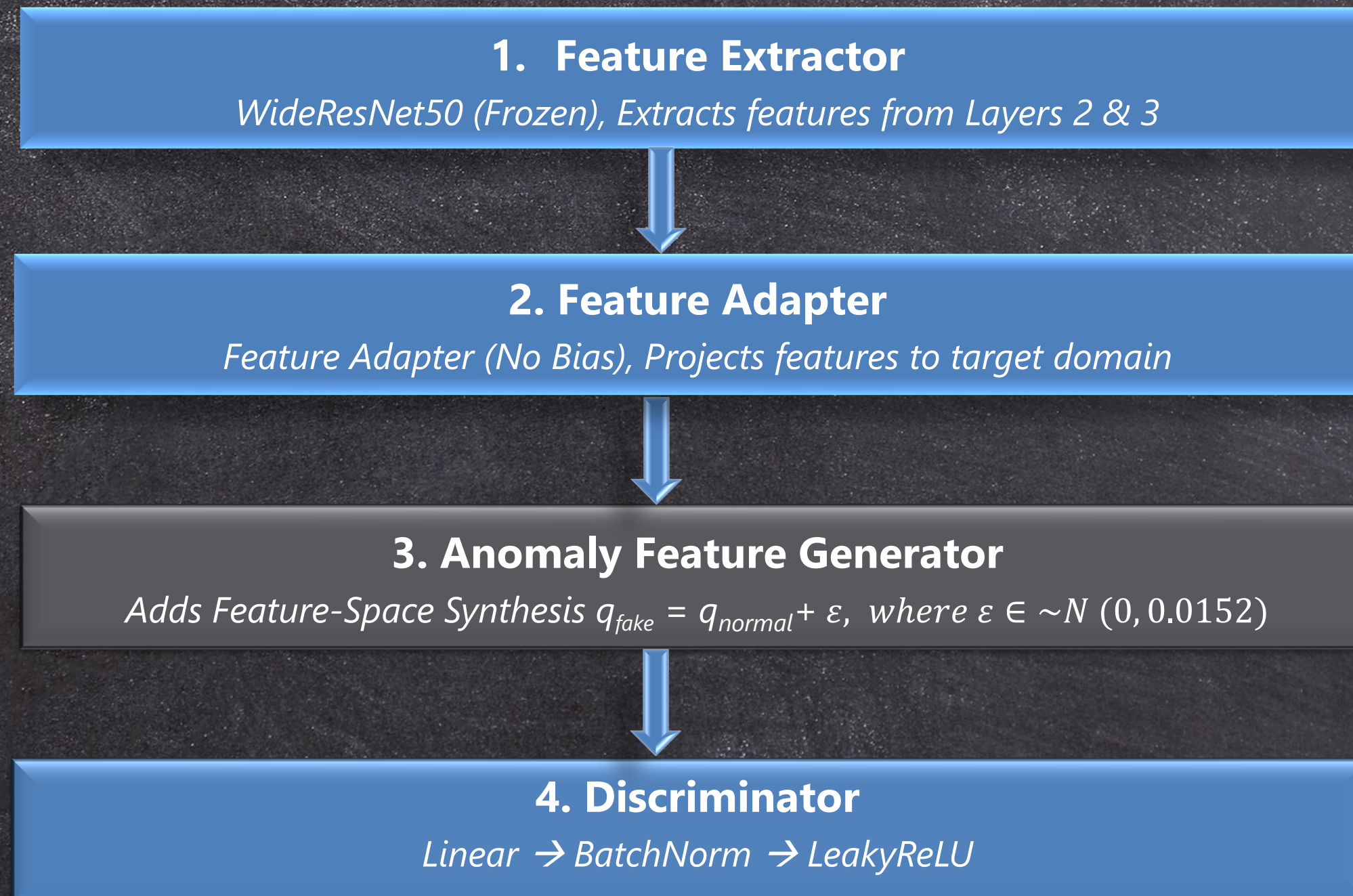
- Scarcity
- Unpredictability
- Real Time

Why Existing Methods Fail?

- × Reconstruction-based
- × Synthesis-based
- × Embedding-based

Why SimpleNet? Simple, Fast and Effective approach

How SimpleNet Works ?



*Key Insight: Synthesize anomalies in **feature space**, not image space!*

SimpleNet Results on MVTec AD

Type	Reconstruction-based		Synthesizing-based		Embedding-based				Ours
Model	AE-SSIM	RIAD	DRÆM	CutPaste	CS-Flow	PaDiM	RevDist	PatchCore	SimpleNet
Carpet	87/64.7	84.2/96.3	97.0/95.5	93.9/98.3	100/-	99.8/99.1	98.9/98.9	98.7/ 99.0	99.7/98.2
Grid	94/84.9	99.6/98.8	99.9/99.7	100/97.5	99.0/-	96.7/97.3	100/99.3	98.2/98.7	99.7/98.8
Leather	78/56.1	100/99.4	100/98.6	100/99.5	100/-	100/99.2	100/99.4	100/99.3	100/99.2
Tile	59/17.5	98.7/89.1	99.6/ 99.2	94.6/90.5	100/-	98.1/94.1	99.3/95.6	98.7/95.6	99.8/97.0
Wood	73/60.3	93.0/85.8	99.1/ 96.4	99.1/95.5	100/-	99.2/94.9	99.2/95.3	99.2/95.0	100/94.5
Avg. Text.	78/56.7	95.1/93.9	99.1/ 97.9	97.5/96.3	99.8/-	95.5/96.9	99.5/97.7	99.0/97.5	99.8/97.5
Bottle	93/83.4	99.9/98.4	99.2/ 99.1	98.2/97.6	99.8/-	99.1/98.3	100/98.7	100/98.6	100/98.0
Cable	82/47.8	81.9/84.2	91.8/94.7	81.2/90.0	99.1/-	97.1/96.7	95.0/97.4	99.5/ 98.4	99.9/97.6
Capsule	94/86.0	88.4/92.8	98.5/94.3	98.2/97.4	97.1/-	87.5/98.5	96.3/98.7	98.1/98.8	97.7/ 98.9
Hazelhut	97/91.6	83.3/96.1	100/99.7	98.3/97.3	99.6/-	99.4/98.2	99.9/98.9	100/98.7	100/97.9
Metal Nut	89/60.3	88.5/92.5	98.7/ 99.5	99.9/93.1	99.1/-	96.2/97.2	100/97.3	100/98.4	100/98.8
Pill	91/83.0	83.8/95.7	98.9/97.6	94.9/95.7	98.6/-	90.1/95.7	96.6/98.2	96.6/97.4	99.0/98.6
Screw	96/88.7	84.5/98.8	93.9/97.6	88.7/96.7	97.6/-	97.5/98.5	97.0/ 99.6	98.1/99.4	98.2/99.3
Toothbrush	92/78.4	100/98.9	100/98.1	99.4/98.1	91.9/-	100/98.8	99.5/ 99.1	100/98.7	99.7/98.5
Transistor	90/72.5	90.9/87.7	93.1/90.9	96.1/93.0	99.3/-	94.4/97.5	96.7/92.5	100/96.3	100/97.6
Zipper	88/66.5	98.1/97.8	100/98.8	99.9/99.3	99.7/-	98.6/98.5	98.5/98.2	99.4/98.8	99.9/ 98.9
Avg. Obj.	91/75.8	89.9/94.3	97.4/97.0	95.5/95.8	98.2/-	96.0/97.8	98/97.9	99.2/ 98.4	99.5/98.4
Average	87/69.4	91.7/94.2	98.0/97.3	96.1/96.0	98.7/-	95.8/97.5	98.5/97.8	99.1/ 98.1	99.6/98.1

Table 1: Comparison of SimpleNet with state-of-the-arts works on MVTec AD

Model	I-AUROC%	P-AUROC%
ResNet18	98.3	95.7
ResNet50	99.6	98.0
ResNet101	99.2	97.6
WideResNet50	99.6	98.1

Table 2: Performanye under different backbones on MVTec AD

Core Idea of SimpleNet

99.6% I-AUROC > (State-of-the-Art on MVTec AD)

Reduces error by 55.5% vs. PatchCore

Reason 1: The Goldilocks Noise

- Optimal Noise ($\sigma = 0.015$)
- Creates a sharp decision boundary without overlapping normal data

Reason 2: Strategic Feature Selection

- Intermediate Layers (2 & 3)
- Captures textures and local geometry instead of high level objects

Implementation: Bridging Industry and Medicine



Task Definition

Trained ONLY on "Normal" scans (1,341 images) to learn the anatomical manifold of a healthy chest.



Backbone: ResNet18

Utilizing Layer 3 features to capture textural information for medical pathologies superior to Layer 4.



Variable Experiment

Noise Scale (σ): 0.015 (Standard) vs. 0.005 (Ablation).

Capacity: 3-layer Discriminator (512 units).



Infrastructure

Batch Size: 16 (Reduced for GPU stability on 11GB VRAM cluster during full dataset runs).

Experimentation Comparison



The RGB Gap

SimpleNet is hard-coded for 3-channel RGB. I modified the data loader to perform 1-to-3 channel expansion for grayscale X-rays.



Computational Cost

Utilized High-Resolution (256x256) processing to capture pneumonia textures, increasing GPU memory overhead.



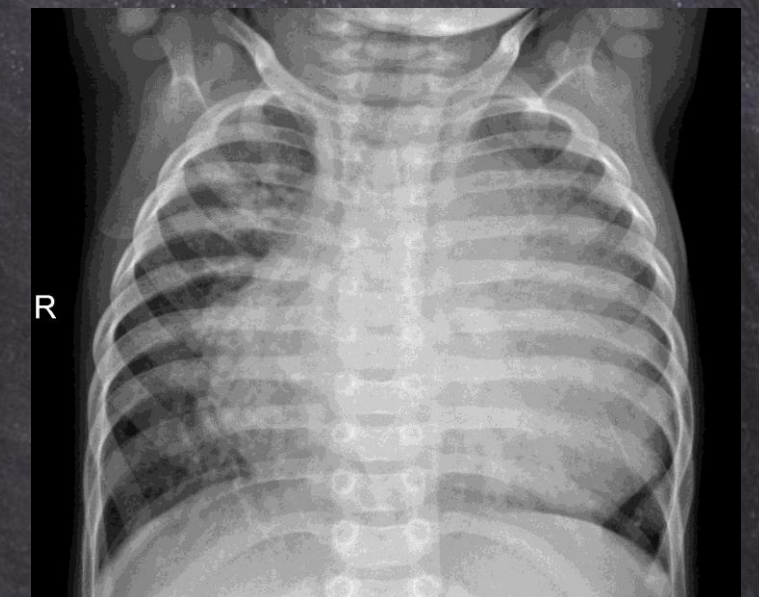
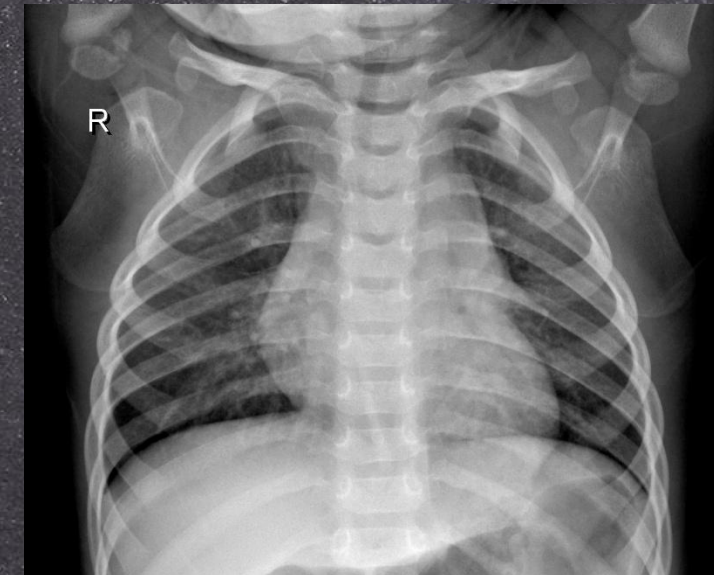
VRAM Stability

Batch size was reduced to 16 to maintain cluster stability on the 11GB VRAM hardware during full-dataset runs.

Implementation: Bridging Industry and Medicine

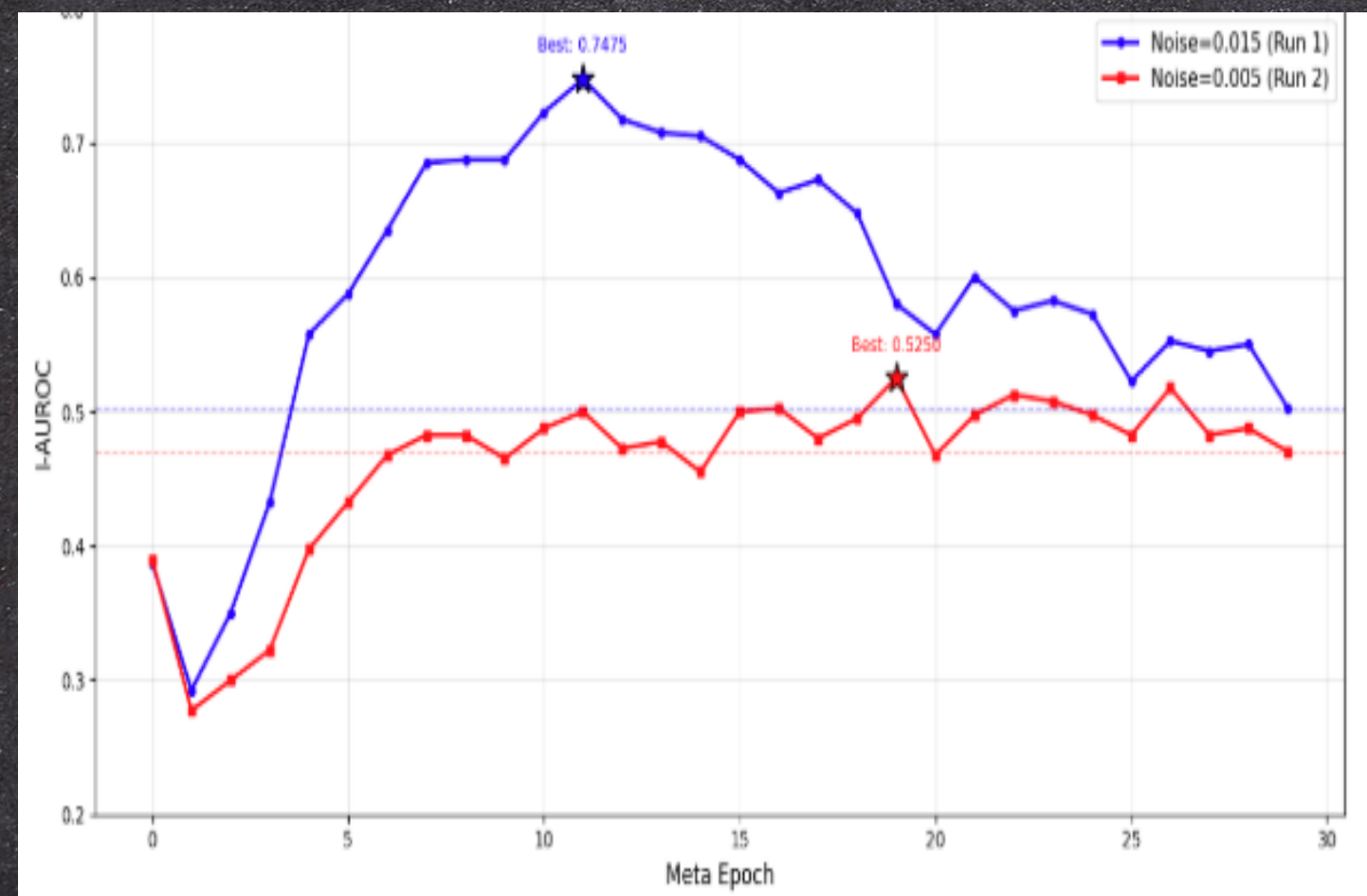
The GroundTruth Problem : MVTec typically expects pixel-level masks for P-AUROC. Our medical dataset provided only image-level labels.

The Solution : Dummy Masks" (Zero-Tensors). This allowed training scripts to focus purely on Image-level AUROC as the success metric

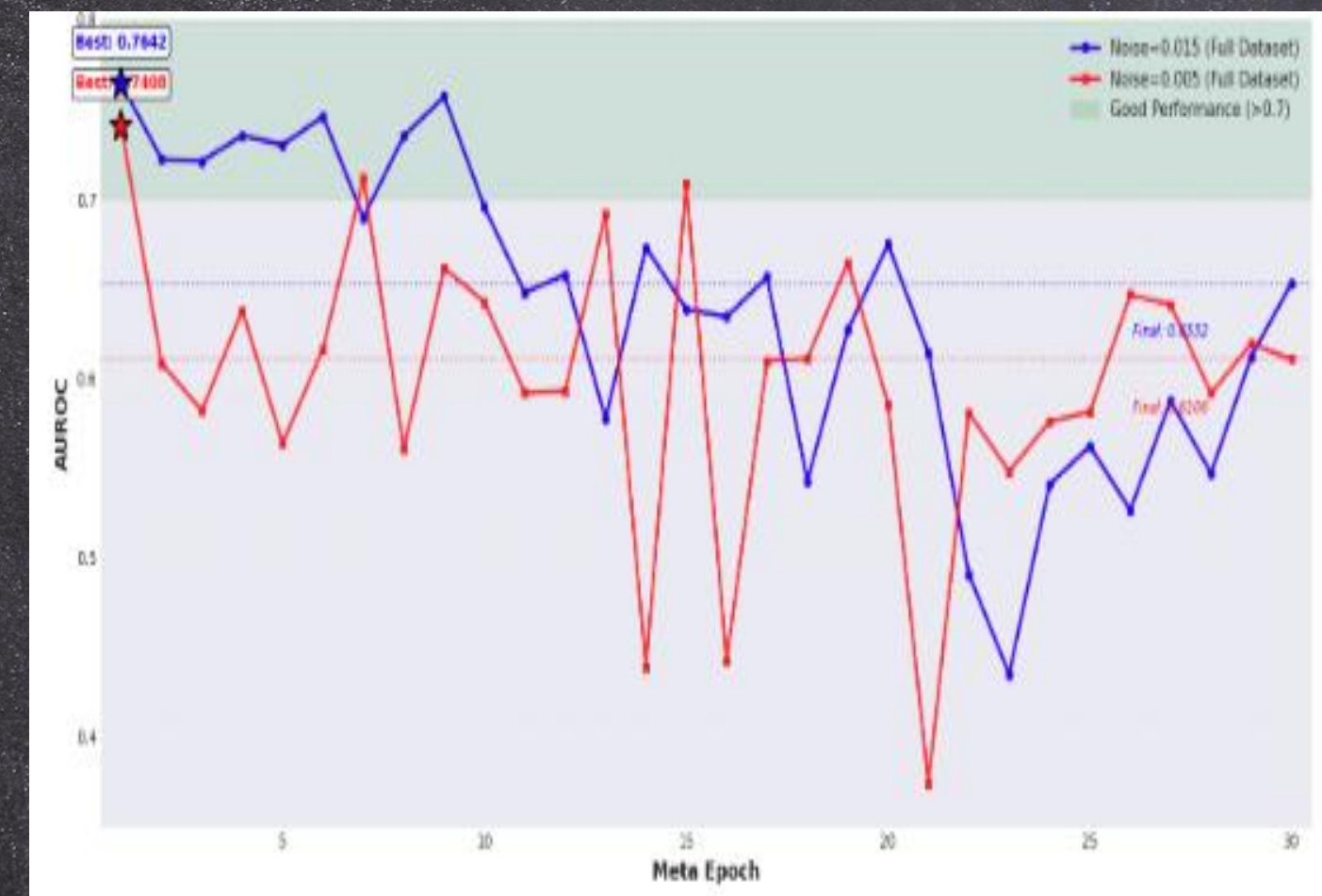


Visual Insight : The Noise Gap

0.764 Best I-AUROC Score



Trained on only 100 normal images with $\sigma = 0.005$ and 0.015



Trained on full dataset around 1340 normal images with $\sigma = 0.005$ and 0.015

Final Verdict : Simplicity is Sophistication

Core Takeaways

- High Versatility: Minimal architectural changes to move from Industry to Medical data
- Rapid Deployment: Clinical triage ready using very few healthy samples.

Future Work

- Adaptive Noise : Based on the Usecase.
- Multi-Modal: Integrating clinical metadata alongside X-ray features.

Thank you 😊 !